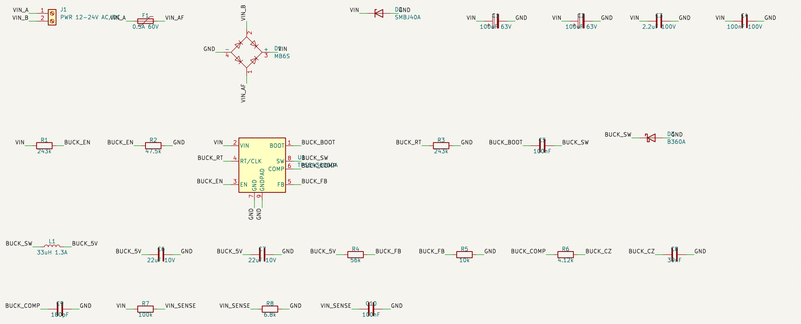
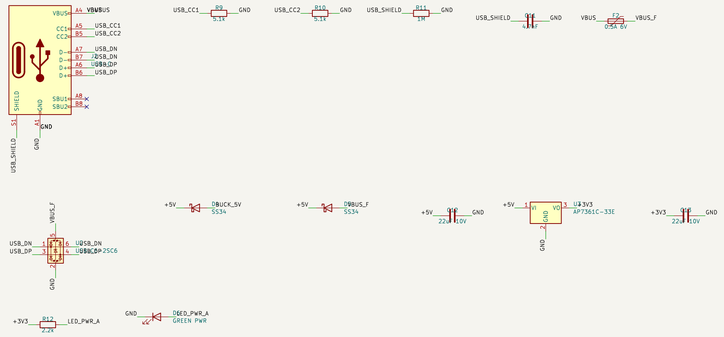
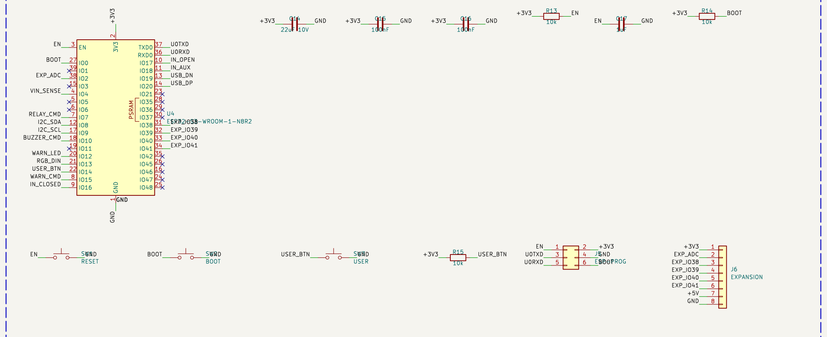
1 POWER INPUT + BUCK



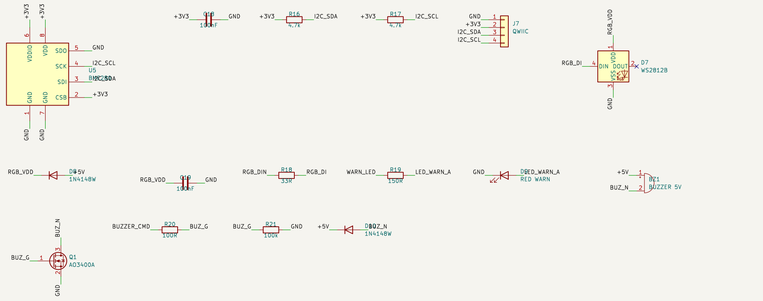
2 USB-C + 5V/3V3 RAILS



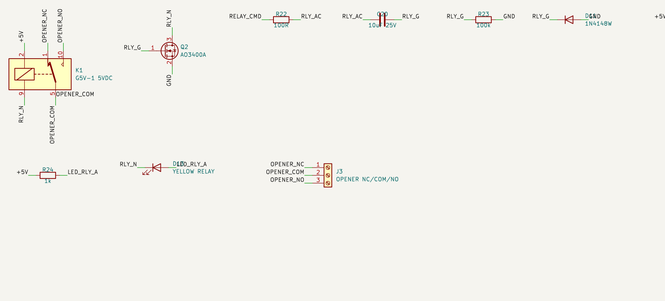
3 ESP32-S3



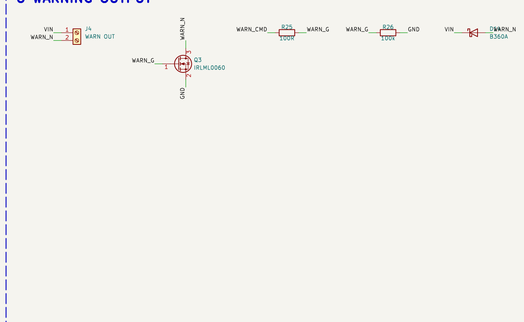
4 SENSORS + UI



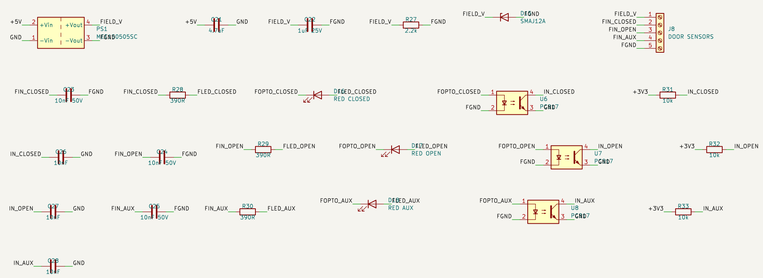
5 RELAY



6 WARNING OUTPUT



7 ISOLATED INPUTS



8 TEST POINTS + MECHANICAL



PWR_FLAG markers (ERC only)



gdc1 rev A — smart garage door controller (retrofit). Relay NO/COM in parallel with the wall button.
Input: 9–36 VDC any polarity or 24 VAC. USB-C 5 V alone also runs the board (bench / flashing).
U1 compensation (R6, C8, C9) and UVLO (R1, R2) are calculated starting values — see docs/DESIGN.md.
C20/R23 make the relay drive AC-coupled: a stuck-high GPIO cannot hold the opener button (SR4).

Isolated: FIELD_V/FGND (door sensors) and OPENER_* (relay contacts). Everything else is logic GND.
Generated from hardware/gen/design.py — edit there and regenerate

Sheet: /	
File: gdc1.kicad_sch	
Title: gdc1 smart garage door controller	
Size: A1	Date: 2026-10-29
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: A**
	Id: 1/1